

TA Class 2

Growth, Consumption-Saving, and Heterogeneity

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Problem Set 2

The Solow Residual Is Bigger Than Technology

The accounting object

$$g_{\hat{A}} = g_Y - \alpha g_K - (1 - \alpha) g_L$$

The residual is whatever output does beyond measured input growth.

(Solow, 1957)

It can contain

- genuine innovation and technology adoption
(Romer, 1990; Comin and Hobijn, 2010)
- capacity utilization, markups, and increasing returns
(Basu and Fernald, 1997; Basu, Fernald, and Kimball, 2006)
- reallocation across firms, sectors, and uses of capital
(Restuccia and Rogerson, 2008; Hsieh and Klenow, 2009)
- institutions, crime, and policy wedges
(Hall and Jones, 1999; Pinotti, 2015)
- input quality and measurement error
(Young, 1995; Hsieh, 2002)

From Solow to Endogenous Growth

Solow Explains Transitions, Not the Source of Long-Run Growth

What Solow explains well

(Solow, 1956)

- Capital deepening during a transition
- Conditional convergence towards a steady state
- How saving, depreciation, and population growth affect levels

What remains outside the model

- On a balanced growth path, growth in output per worker is growth in A
- The model assumes the path of A instead of explaining who creates it
- Sustained growth requires a force that escapes diminishing returns

Endogenous growth asks: who produces A , why, and how does it diffuse?

Three Mechanisms Can Sustain Growth

Mechanism	Why growth persists	What we measure	Research question
AK / human capital (Lucas, 1988)	Broad capital has constant or socially increasing returns	Investment, schooling, skills, learning by doing	Are private and social returns different?
Ideas / Romer (Romer, 1990)	Ideas are non-rival and knowledge spills across users	R&D, researchers, patents, citations, adoption	Who pays for ideas that others can reuse?
Creative destruction (Aghion and Howitt, 1992)	Innovation rents finance quality improvements and reallocation	Entry, exit, markups, firm productivity, patent quality	When does competition encourage innovation?

The unifying tension

Knowledge spillovers create increasing returns; innovation incentives require some excludability or market power.

Different Growth Mechanisms Require Different Data

Cross-country accounting

- Output, capital, employment, human capital, and labour shares
- Compare levels and growth rates, not only GDP
- Useful starting point; weak causal identification

Data

Penn World Table 11.0: 185 countries, 1950–2023.
(Feenstra, Inklaar, and Timmer, 2015)

Innovation and diffusion

- R&D expenditure and researchers
- Patents, citations, and product innovation
- Technology adoption, firm entry and exit

Data

UNESCO/World Bank R&D indicators; WIPO innovation data; firm-level administrative records.

A correlation between R&D and growth does not identify the social return to innovation.

Problem Set 3

Consumption and Saving Beyond the Benchmark

Two Equations Organize the Benchmark

Lifetime resources

$$c_1 + \frac{c_2}{1+r} = a_0 + y_1 - \tau_1 + \frac{y_2 - \tau_2}{1+r}$$

Consumption depends on the present value of resources, not current income alone.

Intertemporal margin

$$u'(c_1) = \beta(1+r)u'(c_2)$$

The interest rate changes the relative price of present and future consumption.

The sharp predictions come from the assumptions:

- free borrowing and lending at a common rate;
- known income, taxes, and government spending;
- lump-sum taxation and a representative household.

The MPC Measures the Response to Additional Resources

Definition

For an unexpected cash transfer x , the marginal propensity to consume in period 1 is

$$\text{MPC}_1 \equiv \frac{\partial c_1}{\partial x},$$

holding the interest rate and future income fixed. It is a response to a shock, not the ratio c_1/y_1 .

Two-period example

$$c_1 + \frac{c_2}{1+r} = W + x, \quad \beta(1+r) = 1 \Rightarrow c_1 = c_2.$$

Formula used below

$$c_1 = \frac{W + x}{1 + (1+r)^{-1}}$$

$$\text{MPC}_1 = [1 + (1+r)^{-1}]^{-1}$$

One extra unit of resources is divided between consumption today and consumption tomorrow.

More Periods Change the Horizon, Not the Logic

Sequential problem

$$c_t + a_{t+1} = y_t - \tau_t + (1 + r_t)a_t$$

Euler equation

$$u'(c_t) = \beta E_t[(1 + r_{t+1})u'(c_{t+1})]$$

What the extra periods add

- a longer horizon over which to spread temporary income;
- expectations about future income and interest rates;
- repeated borrowing constraints;
- precautionary saving under uncertainty;
- age, retirement, and bequest motives.

With a binding borrowing constraint

The Euler equality becomes an inequality: the household would like to move more resources toward the present but cannot.

Interest Rates Reallocate Consumption Across Time

CRRA preferences make the substitution channel explicit

With $u(c) = c^{1-\sigma}/(1-\sigma)$, the Euler equation implies

$$\frac{c_2}{c_1} = [\beta(1+r)]^{1/\sigma}, \quad \text{IES} = \frac{1}{\sigma}.$$

Pure intertemporal substitution

- A higher r raises the reward to postponing consumption
- Holding lifetime resources fixed, c_2/c_1 rises
- Current consumption falls and saving rises

Observed consumption may also reflect

- wealth effects on assets and liabilities
- changes in the present value of future income
- borrowing constraints that limit adjustment

Keep the objects separate

The MPC varies **resources**. The IES governs the response to a change in the **relative price of consumption across dates**.

Ricardian Equivalence Is a Tax-Timing Result

Government intertemporal budget

Holding government spending fixed, a tax cut today financed by debt requires higher taxes in the future.

$$PV(\text{taxes}) = \text{initial debt} + PV(\text{government spending})$$

(Barro, 1974)

Ricardian household

- Present-value tax burden is unchanged
- Consumption is unchanged
- The tax cut is saved to pay future taxes

What the result does not say

- Government spending never matters
- Deficits never affect interest rates
- Distortionary taxes are irrelevant

The proposition is about financing, not the irrelevance of fiscal policy.

Liquidity, Not Just Income, Organizes MPCs

High MPC

- Little liquid wealth
- Close to a borrowing constraint
- High debt service
- Temporary cash relaxes an immediate constraint

Lower MPC

- Large liquid buffer
- Long horizon and stable expected income
- Strong bequest or precautionary motive
- Easy access to credit

Wealthy hand-to-mouth

A household can own a house or pension assets and still have little cash available for current spending. Net worth and liquidity are not the same state variable.

(Kaplan, Violante, and Weidner, 2014)

Household Data Answer Different Questions

Brazil: POF 2017–2018

- Detailed expenditures, income, durables, and household characteristics
- Useful for budget shares and descriptive heterogeneity
- Cross-sectional: not a causal MPC by itself

United States: Consumer Expenditure Survey

- Interview and diary expenditure data
- Public microdata and repeated observations
- Useful for short-run expenditure dynamics

A useful POF exercise

Plot C/Y by income decile, then ask why the slope is not a causal MPC curve and what shock would identify one.

Data: IBGE POF microdata; BLS Consumer Expenditure Survey public microdata.

Heterogeneous-Agent Macro Makes the Distribution a State Variable

Representative-agent model

- One Euler equation
- One common MPC
- Complete risk sharing
- Fiscal transfers largely cancel in aggregate

Heterogeneous-agent model

(Kaplan, Moll, and Violante, 2018; Kaplan and Violante, 2022)

- Distribution of liquid and illiquid wealth
- Idiosyncratic income and employment risk
- Borrowing constraints
- Policy depends on who receives income and when

Why it matters

The same aggregate transfer can generate different consumption responses depending on its size, timing, and distribution across household balance sheets.

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Household data: IBGE POF 2017–2018 microdata; BLS Consumer Expenditure Survey public microdata.

Growth data: Penn World Table 11.0; World Bank and UNESCO R&D indicators; WIPO Global Innovation Index.